Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1.	*(a)*	(i)	The current [through a metal conductor at constant temperature] is proportional to the pd [across it] Accept $I \propto V$ with well-defined symbols.	1			1		
		(ii)	Constant [or independent of current [or pd]] (1)	1			1		
	(b)	(i)	pd across X = 0.70 [V] (1) or by impl so R_X = 58 [Ω] (1) **or** total resistance = 183 [Ω] (1) or by impl so R_X = 58 [Ω] (1)		2		2	2	
		(ii)	R_X = 16 [Ω] or equivalent (1) So Ohm's law not obeyed (1) [not freestanding] **no ecf** **or** If R_X stayed at 58 [Ω] (or 58.3 Ω), then I = 27 (or 26) mA (1) Not so, therefore Ohm's law not obeyed (1) **no ecf**			2	2	1	
		(iii)	No. Filament resistance increases with increasing pd [or current] **or** equivalent **ecf** on calculated value of R_X in (b)(ii)			1	1		
	(c)	(i)	Temperature at which a material [that is classed as a superconductor] loses all its resistivity [accept resistance] [when cooled]. Accept "... below which a material [...] has no resistivity [accept resistance]"	1			1		
		(ii)	Coil for MRI machine, power cables, magnets / electromagnets or anything reasonable (1) Can be kept below transition temp [cheaply] by liquid nitrogen [accept: superconducts in liquid nitrogen] (1)	2			2		
			Question 1 total	**5**	**2**	**3**	**10**	**3**	**0**